



University of the East
Manila Campus
College of Engineering
Computer Engineering Department



NCP4304: Embedded Systems 2 Laboratory
(Internet of Things and Data Analytics Track)
1st Semester, Academic Year 2025 - 2026

Exploratory Data Analysis

Laboratory Work No. 5

Student Name:

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Surname (UPPERCASE) Given Name (UPPERCASE) M.I.

Student Name:

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Date
Performed:

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Date
Submitted:

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Section:

CPE 1

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Facilitator:

Engr. Errol John M. Antonio

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Grade:

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Execution of the
Experiment / Exercise

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Laboratory Report

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Total



Disclosure

I/We acknowledge that I/we have followed the university and department guidelines on the ethical use of Generative AI tools in completing this submission.

Please check (✓):

☐

I/We did not use any AI tools in the preparation of this work.

☐

I/We used generative AI tools in the following manner:

Generative AI Tool	Purpose

I/We affirm that the final content is my/our own work, and any use of AI tools did not involve the generation of fabricated data, false references, or copying of AI-generated content without proper review and attribution. I/We understand that violations may result in sanctions and the nullification of this submission.

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Signature over Printed Name

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Signature over Printed Name



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Grading Rubric for the Conduct of Laboratory Exercise / Experiment

Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Impr. (0-2)	Score
Execution and Output of the Experiment (x3)	Demonstrated full adherence to procedures, correct setup, and effective execution of the experiment. Showed initiative and independence in troubleshooting and optimizing the process.	Followed procedures correctly with minimal guidance. Minor errors in setup or execution that did not affect results significantly.	Needed frequent assistance or had difficulty setting up. Execution had issues affecting the outcomes.	Did not follow procedures. Experiment not completed, improperly executed, or results invalid due to errors.	/15
Quality of Output (x5)	The output is of high quality : fully functional, stable, and complete. It performs all required functions without errors, is neatly implemented, and exceeds expectations in precision or efficiency.	The output is functional and mostly complete , with minor issues that do not significantly affect performance. The work shows attention to detail and is mostly aligned with the intended goal.	The output is partially working or incomplete , with noticeable issues or instability. Some requirements were met, but the performance or accuracy is below expectations.	The output is non-functional, unstable, or far from expectations . Major components are missing or do not perform as intended. Work appears rushed or poorly executed.	/25
Timeliness of Completion (x2)	Completed before the deadline with no extensions needed.	Completed on deadline with no extensions needed.	Completed slightly after the deadline with a valid reason or minimal delay.	Completed late with noticeable delay or required multiple reminders.	/10
Bonus(+): Award for exceptional effort, overcoming significant challenges, or additional valuable features. Penalties(-): Deduct points for lack of professionalism, poor adherence to guidelines, or missed deadlines.					Total
Remarks:					



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Grading Rubric for Laboratory Design Experiment Report

Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Impr. (0-2)	Score
Answers to Questions (x2)	Answers are thorough and accurate , demonstrating a deep understanding of the experiment.	Answers are accurate and show a good understanding , with minor inaccuracies.	Answers are incomplete or show a basic understanding , with some inaccuracies.	Answers are inaccurate or incomplete , showing little understanding .	/10
Observations and Discussion (x3)	Observations are detailed and insightful , and the discussion is well-reasoned, linking results to the experiment's objectives.	Observations are clear , and the discussion is logical , with minor gaps in reasoning.	Observations are basic , and the discussion is superficial or lacks depth.	Observations are unclear or missing , and the discussion is incomplete or illogical .	/15
Conclusion (x3)	The conclusion is clear , well-supported by the data, and effectively summarizes the experiment with insights, implications , and recommendations .	The conclusion is clear and supported by the data, with minor gaps. Summarize key results but may not fully link them to the experiment's objectives.	The conclusion is basic and partially supported by the data. Summarizes results, but may fail to connect them to the main objectives .	The conclusion is unclear, unsupported by the data, or missing. Results are not summarized , or there is little connection to the experiment's objectives.	/15
Writing Quality (x2)	The report is written with excellent grammar, spelling, and sentence structure. The language is precise, and ideas are communicated effectively without ambiguity.	The report is well-written , with only minor grammatical, spelling, or structural errors. The ideas are mostly clear and well-expressed	The report has several grammatical, spelling, or structural errors that occasionally hinder understanding. Some ideas may be unclear or poorly expressed.	The report is poorly written , with frequent grammatical, spelling, or structural errors, making it difficult to understand. Ideas are poorly communicated or unclear.	/10
Presentation (x2)	The report is well-organized , with clear headings, consistent formatting, and no errors.	The report is organized and mostly error-free , with minor formatting inconsistencies.	The report is somewhat organized , with noticeable formatting errors and inconsistencies.	The report is poorly organized , with significant formatting errors and inconsistencies.	/10
Bonus(+): Award for exceptional effort, overcoming significant challenges, or additional valuable features. Penalties(-): Deduct points for lack of professionalism, poor adherence to guidelines, or missed deadlines.					Total
Remarks:					



Exploratory Data Analysis

Laboratory Work No. 5

Intended Learning Outcomes

By the end of this laboratory experiment/exercise, students should be able to:

- Identify and apply various methods of the Pandas library used for exploratory data analysis (EDA).
- Perform data preprocessing, cleaning, and manipulation techniques to prepare IoT datasets for analysis using the Pandas library.
- Interpret patterns and trends in the dataset by applying summary statistics and exploratory methods.
- Utilize appropriate data visualization techniques (e.g., bar charts, histograms, heatmaps) to effectively communicate insights from IoT data.

Overview

In the Internet of Things (IoT) ecosystem, large volumes of data are continuously collected from various interconnected sensors and devices. Before advanced modeling or analytics can be applied, it is crucial to understand the structure, quality, and behavior of this data. Exploratory Data Analysis (EDA) plays a key role in this process.

This laboratory experiment introduces students to the fundamentals of EDA using Python's Pandas library. Students will learn how to load, examine, clean, and manipulate raw IoT datasets. Through statistical summaries and visualization techniques, they will uncover patterns, detect anomalies, and generate insights that are essential for decision-making and system optimization.

The laboratory experiment also emphasizes the use of basic visualization tools (such as histograms, line plots, bar charts, and heatmaps) to support data-driven interpretations. By the end of this session, students will be equipped with practical skills in performing EDA that can be applied to real-world IoT applications, such as environmental monitoring, smart agriculture, and industrial automation.

Please refer to this supplementary Google Collaboratory notebook.



Dataset Needed

- `titanic_dataset.csv`
- `lot_env_data.csv`

Dataset Structure:

- `timestamp` – Date and time of the reading
- `sensor_id` – Unique identifier of the sensor
- `sensor_type` – Type of sensor (e.g., temperature, humidity)
- `location` – Physical location of the sensor]
- `temperature` – Measured in °C
- `humidity` – Relative humidity in %
- `air_quality` – Air Quality Index (0–500)
- `light` – Light intensity in lux



Tools Needed:

- Python IDE (Google Colaboratory / Anaconda - Jupyter)
- Libraries: pandas, matplotlib, seaborn



Tasks:

Provide the necessary scripts to obtain the needed data/information in each item. Also, a snapshot of your source code and output on the space provided for each item.

Part 1: Understanding the IoT Dataset

1. Import all necessary libraries for handling and visualizing data.
2. Read the dataset `iot_env_data.csv` and display the first five rows.
3. Use the appropriate method to display the number of entries, data types, and null values.
4. Identify which columns have missing data and how many are missing in each.
5. Compute the average values for:
 - a. Temperature
 - b. Humidity
 - c. Air Quality
 - d. Light Intensity



Part 2: Patterns and Trends

6. Determine whether the **morning (AM)** or **afternoon (PM)** has a higher **average temperature**. Display the results using a bar graph.
7. Identify which sensor device (based on **sensor_id**) has the **most readings recorded**.
8. Create a **line graph** to show how the **temperature varies over time**. Observe and describe any visible trends.
9. Identify the **top 5 dates** with the highest average air quality readings.
10. Generate a **heatmap** to show correlations between numerical sensor values (e.g., temperature, humidity, air quality, light).

Part 3: Sensor Behavior and Anomalies

11. Use a **boxplot** to examine whether any outliers exist in:
 - a. Temperature
 - b. Humidity
12. Count the number of **unique sensor types** and **unique** locations in the dataset.
13. Determine the **top 5 most active sensor locations** based on the frequency of recorded data. Display the results using a bar chart.
14. Plot a **histogram** showing the distribution of air quality readings.
15. Investigate how light intensity changes over the course of the day. Present the average light intensity per hour using a line graph.

Attach a copy of your complete and organized Google Collaboratory notebook. Ensure that **"text"** and code **"comments"** are properly placed.

Congratulations! You did great on your Laboratory Work!



 Questions to Answer:

1. What is the purpose of performing exploratory data analysis (EDA) before applying advanced analytics to IoT data?

2. Why is it important to visualize IoT sensor data instead of relying solely on raw numbers and summary statistics?

3. What insights can you draw from a correlation heatmap of temperature, humidity, air quality, and light intensity data collected from IoT sensors?



4. How can missing or inconsistent sensor data impact the results of your analysis, and what steps can you take to address it?

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5. Why is it important to group and analyze IoT data based on time (e.g., hourly, daily)? What kind of trends might this reveal?

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 **Observations & Discussion**

 **Conclusion**

 **References**

 **Appendices**



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Observations & Discussion:



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 **Conclusion:**